ETALE LOCAL STRUCTURE OF A FAMILY OF CURVES AT A SINGULAR POINT OF THE FIBRE: ETALE PULLBACK TO THE STANDARD UV - A FAMILY OVER Z[A]

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the approximation reduction with its base-change compatibility of Sing(f), the etale neighbourhood realizing kappa'/kappa(s) in both the split and field cases, the completed normal form st - h with the explicit fibre product presentation of Y_sigma, and the closing Artin approximation step are all retained verbatim.

Editorial note. The label lemma-etale-local-structure-nodal-family names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Absolute Noetherian approximation. Limits, Lemma limits-lemma-descend-finite-presentation, with Lemma lemma-descend-nodal-family and Divisors, Lemma divisors-lemma-base-change-and-fitting-ideal-omega. Supplies the reduction to S of finite type over Z: write R = colim R_i, descend f to f_i at-worst-nodal of relative dimension 1 with x_i a singular point of the fibre.
- Artin approximation for the completed local algebra. More on Morphisms, Lemma more-morphisms-lemma-relative-map-approximation-pre, with More on Algebra, Proposition more-algebra-proposition-ubiquity-G-ring. Turns the completed isomorphism O_{X,x}^wedge = O_{Y_sigma,y_sigma}^wedge into an etale-local isomorphism after an etale neighbourhood, since O_{S,s} is a G-ring.

Source reference. Original reference 0CBV: sources/stacks-curves/pinned-original-source.tex, source lines 4572–4572. Prerequisite pointer invoked at proof line(s) 5356; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C3I: sources/stacks-chow/tags.txt, source lines 15193–15193. Prerequisite pointer invoked at proof line(s) 5329; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C5F: sources/stacks-curves/pinned-original-source.tex, source lines 5096–5096. Prerequisite pointer invoked at proof line(s) 5325; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBX: sources/stacks-curves/pinned-original-source.tex, source lines 5137–5137. Prerequisite pointer invoked at proof line(s) 5357; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C4D: sources/stacks-curves/pinned-original-source.tex, source lines 4391–4391. Prerequisite pointer invoked at proof line(s) 5340; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Prerequisite pointer invoked at proof line(s) 5321; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07PX: sources/stacks-chow/tags.txt, source lines 9812–9812. Prerequisite pointer invoked at proof line(s) 5384; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02LF: sources/stacks-chow/tags.txt, source lines 3406–3406. Prerequisite pointer invoked at proof line(s) 5345; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GDX: sources/stacks-chow/tags.txt, source lines 20457–20457. Prerequisite pointer invoked at proof line(s) 5386; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBT: sources/stacks-curves/pinned-original-source.tex, source lines 4521–4521. Prerequisite pointer invoked at proof line(s) 5343; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume that f is at-worst-nodal of relative dimension 1. Let $x \in X$ be a point which is a singular point of the fibre X_s. Then there exists a commutative diagram of schemes*

$$\begin{array}{ccccccc} X & \leftarrow & U & & \to & W & \to \operatorname{Spec}(\mathbf{Z}[u,v,a]/(uv-a)) \\ \downarrow & & & \searrow & \swarrow & & \downarrow \\ S & \leftarrow & & V & & \to & \operatorname{Spec}(\mathbf{Z}[a]) \end{array}$$

with $X \leftarrow U$, $S \leftarrow V$, and $U \to W$ étale morphisms, and with the right hand square cartesian, such that there exists a point $u \in U$ mapping to x in X.

First proof using Artin approximation. We first use absolute Noetherian approximation to reduce to the case of schemes of finite type over $\mathbf{Z}$. The question is local on X and S. Hence we may assume that X and S are affine. Then we can write $S = \operatorname{Spec}(R)$ and write R as a filtered colimit $R = \operatorname{colim} R_i$ of finite type $\mathbf{Z}$-algebras. Using Limits, Lemma 01ZM we can find an i and a morphism

$f_i : X_i \to \mathrm{Spec}(R_i)$ whose base change to S is f. After increasing i we may assume that f_i is at-worst-nodal of relative dimension 1, see Lemma 0C5F. The image $x_i \in X_i$ of x will be a singular point of its fibre, for example because the formation of $\mathrm{Sing}(f)$ commutes with base change (Divisors, Lemma 0C3I). If we can prove the lemma for $f_i : X_i \to S_i$ and x_i, then the lemma follows for $f : X \to S$ by base change. Thus we reduce to the case studied in the next paragraph.

Assume S is of finite type over $\mathbf{Z}$. Let $s \in S$ be the image of x. Recall that $\kappa(x)$ is a finite separable extension of $\kappa(s)$, for example because $\mathrm{Sing}(f) \to S$ is unramified or because x is a node of the fibre X_s and we can apply Lemma 0C4D. Furthermore, let $\kappa'/\kappa(x)$ be the degree 2 separable algebra associated to $\mathcal{O}_{X_s,x}$ in Remark 0CBT. By More on Morphisms, Lemma 02LF we can choose an étale neighbourhood $(V, v) \to (S, s)$ such that the extension $\kappa(v)/\kappa(s)$ realizes either the extension $\kappa(x)/\kappa(s)$ in case $\kappa' \cong \kappa(x) \times \kappa(x)$ or the extension $\kappa'/\kappa(s)$ if κ' is a field. After replacing X by $X \times_S V$ and S by V we reduce to the situation described in the next paragraph.

Assume S is of finite type over $\mathbf{Z}$ and $x \in X_s$ is a split node, see Definition 0CBV. By Lemma 0CBX we see that there exists an $\mathcal{O}_{S,s}$-algebra isomorphism

$$\mathcal{O}^\wedge_{X,x} \cong \mathcal{O}^\wedge_{S,s}[[s, t]]/(st - h)$$

for some $h \in \mathfrak{m}^\wedge_s \subset \mathcal{O}^\wedge_{S,s}$. In other words, if we consider the homomorphism

$$\sigma : \mathbf{Z}[a] \longrightarrow \mathcal{O}^\wedge_{S,s}$$

sending a to h, then there exists an $\mathcal{O}_{S,s}$-algebra isomorphism

$$\mathcal{O}^\wedge_{X,x} \longrightarrow \mathcal{O}^\wedge_{Y_\sigma, y_\sigma}$$

where

$$Y_\sigma = \mathrm{Spec}(\mathbf{Z}[u, v, t]/(uv - a)) \times_{\mathrm{Spec}(\mathbf{Z}[a]),\sigma} \mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$$

and y_σ is the point of Y_σ lying over the closed point of $\mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$ and having coordinates u, v equal to zero. Since $\mathcal{O}_{S,s}$ is a G-ring by More on Algebra, Proposition 07PX we may apply More on Morphisms, Lemma 0GDX to conclude. □